

1 Question (15 points)

Suppose a dishonest dealer has two coins, one fair and one biased; the biased coin has heads probability $1/4$. Assume that the dealer never switches the coins. Which coin is more likely to have generated the sequence HTTTHH-HTTTTHHTT? ($\log_2(3) = 1.585$).

2 Question (15 points)

Find the optimal pairwise global alignment of the sequences TACGAG**T**ACGA and ACTGAC**G**ACTGAC with the condition that **G** nucleotides shown in bold font must be aligned together. The scoring parameters are defined as +2 for match, -1 for mismatch, and $d = -2$ for a gap penalty.

3 Question (15 points)

Given the following 6 DNA strings,

agAAA

gATGt

aGGCt

ACGgg

aaGGT

where the random motif (randomly selected 3-mers) is given in uppercase form. Run randomized motif search algorithm with these settings 1 iteration. DO NOT USE Laplace smoothing.

4 Question (15 points)

Given the integer masses of 20 amino acids as an hash map (dictionary)

$$G \rightarrow 57$$

$$A \rightarrow 71$$

$$S \rightarrow 87$$

...

and a cyclic peptide such as NQEL (given as string), write a method to generate the theoretical spectrum of the peptide. Your algorithm will take n and d as parameters.

5 Question (20 points)

Write an algorithm that iterate over every index from $(0, 0, \dots, 0)$ to $(n, n, \dots, n)$, where there are d indexes. DO NOT USE any external libraries.

6 Question (20 points)

The string reconstruction problem (SRP) problem is to reconstruct a DNA sequence from its k -mer composition. Suppose that instead of a single target DNA fragment, we have two target DNA fragments. Give a precise formulation of the resulting problem and also give an approach to the above problem which resembles the Eulerian Cycle or Hamiltonian Path approach to SRP.